FROBENIUS-PERIODIC VECTOR BUNDLES GLUE ALONG THE ALMOST BLOWUP SQUARE OF AN EQUATION IN TWO GENERATORS, GIVING AN EQUIVALENCE OF CATEGORIES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; no macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0EXD (sources/stacks-flat/pinned-original-source.tex line 12303).

Allowed standard prerequisites (editorial).

- The almost blowup square associated with an equation in two generators, and its Frobenius properties. Same chapter, flat.tex, Example labelled example-two-generators (source line 11488), invoked in the statement and at proof lines 12338 and 12346. Supplies the square X, X', Z, E and the description X = Spec(A), Z = Spec(A/(f_1, f_2)), X' = Proj(A[T_0, T_1]/(f_2T_0 - f_1T_1)), E = P^1_Z used at proof lines 12366-12367.
- Colimits of F-periodic vector bundles are sheaves and glue in the Zariski topology. Same chapter, Lemmas labelled lemma-colim-F-Vect (source line 12223) and lemma-h-sheaf-colim-F (source line 11855), invoked at proof lines 12346 and 12382. Supply that the colimit category of vector bundles is identified with finite locally free modules over the colimit of the structure sheaves, so that the fibre product of global section colimits computes the maps, and that objects and morphisms may be glued Zariski locally.
- Descent of a triple of finite locally free modules along a finitely generated subalgebra, vanishing of H^1 for globally generated modules on P^1, and the theorem on formal functions. Limits, Lemma labelled limits-lemma-descend-invertible-modules (Stacks tag 0B8W) at proof lines 12368-12372; More on Morphisms, Lemma labelled more-morphisms-lemma-globally-generated-vanishing (Stacks tag 0E7H) at proof lines 12405-12406; Cohomology of Schemes, Theorem labelled coherent-theorem-formal-functions (Stacks tag 02OC) at proof lines 12420-12421. Supply the Noetherian reduction, the vanishing of the first cohomology of a globally generated module on E, and the lifting of the compatible trivialisation on E to a global map on X'; the gluing of the resulting local solutions is inside the counted span.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original definition 0EVH: sources/stacks-flat/pinned-original-source.tex, source lines 11488–11488. Same-chapter pointer into the author environment example at source line 11488; invoked at proof line 12338 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0EVK: sources/stacks-flat/pinned-original-source.tex, source lines 11855–11855. Same-chapter pointer into the author environment lemma at source line 11855; invoked at proof line 12346 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B8W: sources/stacks-chow/tags.txt, source lines 14154–14154. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12369 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0EXB: sources/stacks-flat/pinned-original-source.tex, source lines 12223–12223. Same-chapter pointer into the author environment lemma at source line 12223; invoked at proof line 12382 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0E7H: sources/stacks-chow/tags.txt, source lines 17782–17782. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12406 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original theorem 02OC: sources/stacks-chow/tags.txt, source lines 3511–3511. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12421 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let p be a prime number. Consider an almost blowup square X, X', Z, E in characteristic p as in Example 0EVH. Then the functor*

$$G : \mathrm{colim}_F \mathit{Vect}(X) \longrightarrow \mathrm{colim}_F \mathit{Vect}(Z) \times_{\mathrm{colim}_F \mathit{Vect}(E)} \mathrm{colim}_F \mathit{Vect}(X')$$

is an equivalence.

Proof. Fully faithfulness. Suppose that $(\mathcal{E}, n)$ and $(\mathcal{F}, m)$ are objects of $\mathrm{colim}_F \mathit{Vect}(X)$. Let $(a, b) : G(\mathcal{E}, n) \to G(\mathcal{F}, m)$ be a morphism in the RHS. We may choose $N \gg 0$ and think of a as a map $a : F^{N-n,*}\mathcal{E}|_Z \to F^{N-m,*}\mathcal{F}|_Z$ and b as a map $b : F^{N-n,*}\mathcal{E}|_{X'} \to F^{N-m,*}\mathcal{F}|_{X'}$ agreeing over E. Choose a finite affine open covering $X = X_1 \cup \ldots \cup X_n$ such that $\mathcal{E}|_{X_i}$ and $\mathcal{F}|_{X_i}$ are finite free $\mathcal{O}_{X_i}$-modules. For each i the base change

$$\begin{array}{ccc} E_i & \longrightarrow & X_i' \\ \downarrow & & \downarrow \\ Z_i & \longrightarrow & X_i \end{array}$$

is another almost blow up square as in Example 0EVH. For these squares we know that

$$\mathrm{colim}_F H^0(X_i, \mathcal{O}_{X_i}) = \mathrm{colim}_F H^0(Z_i, \mathcal{O}_{Z_i}) \times_{\mathrm{colim}_F H^0(E_i, \mathcal{O}_{E_i})} \mathrm{colim}_F H^0(X_i', \mathcal{O}_{X_i'})$$

by Lemma 0EVK (see proof of the lemma). Hence after increasing N we may assume the maps $a|_{Z_i}$ and $b|_{X_i'}$ come from maps $c_i : F^{N-n,*}\mathcal{E}|_{X_i} \to F^{N-m,*}\mathcal{F}|_{X_i}$.

After possibly increasing N we may assume c_i and c_j agree on $X_i \cap X_j$. Thus these maps glue to give the desired morphism $(\mathcal{E}, n) \to (\mathcal{F}, m)$ in the LHS.

Essential surjectivity. Let $(\mathcal{F}, \mathcal{G}, \varphi)$ be a triple consisting of a finite locally free $\mathcal{O}_Z$-module $\mathcal{F}$, a finite locally free $\mathcal{O}_{X'}$-module $\mathcal{G}$, and an isomorphism $\varphi : \mathcal{F}|_E \to \mathcal{G}|_E$. We have to show that after replacing this triple by a Frobenius power pullback, it comes from a finite locally free $\mathcal{O}_X$-module.

Noetherian reduction; we urge the reader to skip this paragraph. Recall that $X = \mathrm{Spec}(A)$ and $Z = \mathrm{Spec}(A/(f_1, f_2))$, $X' = \mathrm{Proj}(A[T_0, T_1]/(f_2T_0 - f_1T_1))$, and $E = \mathbf{P}^1_Z$. By Limits, Lemma 0B8W we can find a finitely generated $\mathbf{F}_p$-subalgebra $A_0 \subset A$ containing f_1 and f_2 such that the triple $(\mathcal{F}, \mathcal{G}, \varphi)$ descends to $X_0 = \mathrm{Spec}(A_0)$ and $Z_0 = \mathrm{Spec}(A_0/(f_1, f_2))$, $X'_0 = \mathrm{Proj}(A_0[T_0, T_1]/(f_2T_0 - f_1T_1))$, and $E_0 = \mathbf{P}^1_{Z_0}$. Thus we may assume our schemes are Noetherian.

Assume X is Noetherian. We may choose a finite affine open covering $X = X_1 \cup \ldots \cup X_n$ such that $\mathcal{F}|_{Z \cap X_i}$ is free. Since we can glue objects of $\mathrm{colim}_F \mathit{Vect}(X)$ in the Zariski topology (Lemma 0EXB), and since we already know fully faithfulness over X_i and $X_i \cap X_j$ (see first paragraph of the proof), it suffices to prove the existence over each X_i. This reduces us to the case discussed in the next paragraph.

Assume X is Noetherian and $\mathcal{F} = \mathcal{O}_Z^{\oplus r}$. Using φ we get an isomorphism $\mathcal{O}_E^{\oplus r} \to \mathcal{G}|_E$. Let $I = (f_1, f_2) \subset A$. Let $\mathcal{I} \subset \mathcal{O}_{X'}$ be the ideal sheaf of E; it is globally generated by f_1 and f_2. For any n there is a surjection

$$(\mathcal{I}^n/\mathcal{I}^{n+1})^{\oplus r} = \mathcal{I}^n/\mathcal{I}^{n+1} \otimes_{\mathcal{O}_E} \mathcal{G}|_E \longrightarrow \mathcal{I}^n\mathcal{G}/\mathcal{I}^{n+1}\mathcal{G}$$

Hence the first cohomology group of this module is zero. Here we use that $E = \mathbf{P}^1_Z$ and hence its structure sheaf and in fact any globally generated quasi-coherent module has vanishing H^1. Compare with More on Morphisms, Lemma 0E7H. Then using the short exact sequences

$$0 \to \mathcal{I}^n\mathcal{G}/\mathcal{I}^{n+1}\mathcal{G} \to \mathcal{G}/\mathcal{I}^{n+1}\mathcal{G} \to \mathcal{G}/\mathcal{I}^n\mathcal{G} \to 0$$

and induction, we see that

$$\lim H^0(X', \mathcal{G}/\mathcal{I}^n\mathcal{G}) \to H^0(E, \mathcal{G}|_E) = H^0(E, \mathcal{O}_E^{\oplus r}) = A/I^{\oplus r}$$

is surjective. By the theorem on formal functions (Cohomology of Schemes, Theorem 02OC) this implies that

$$H^0(X', \mathcal{G}) \to H^0(E, \mathcal{G}|_E) = H^0(E, \mathcal{O}_E^{\oplus r}) = A/I^{\oplus r}$$

is surjective. Thus we can choose a map $\alpha : \mathcal{O}_{X'}^{\oplus r} \to \mathcal{G}$ which is compatible with the given trivialization of $\mathcal{G}|_E$. Thus α is an isomorphism over an open neighbourhood of E in X'. Thus every point of Z has an affine open neighbourhood where we can solve the problem. Since $X' \setminus E \to X \setminus Z$ is an isomorphism, the same holds for points of X not in Z. Thus another Zariski glueing argument finishes the proof. $\square$